

Title: **Personalized, Real-Time Health Impact of Climate Change and Pollution**

1. General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name	Institution*	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant/ Co-applicant]
Alina Rwei	TUD	Chemical Engineering	a.y.rwei@tudelft.nl	Tenure-track		Main Lead, WP2 Lead
Pilar Garcia-Gomez	EUR	Applied Economics	garciaagomez@ese.eur.nl	Tenured		Lead, WP3 Lead
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Massimo Mastrangeli	TUD	Microelectronics	m.mastrangeli@tudelft.nl	Tenured		Applicant, WP1 Lead
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Chirlmin Joo	TUD	Bionanoscience	c.joo@tudelft.nl	Tenured		Applicant
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* TUD = Delft University of Technology, EUR = Erasmus University Rotterdam, EMC = Erasmus Medical Centre

Corresponding Lead: Alina Rwei

Planned start date of seed funding: 01/09/2022

Primary Convergence theme: *Human Centered Technology and AI for Health*

Keywords: *wearable bioelectronics, immune response, cytokines, personalized health, organ-on-a-chip, environmental stress, evidenced-based interventions*

1.b Summary (max 300 words)

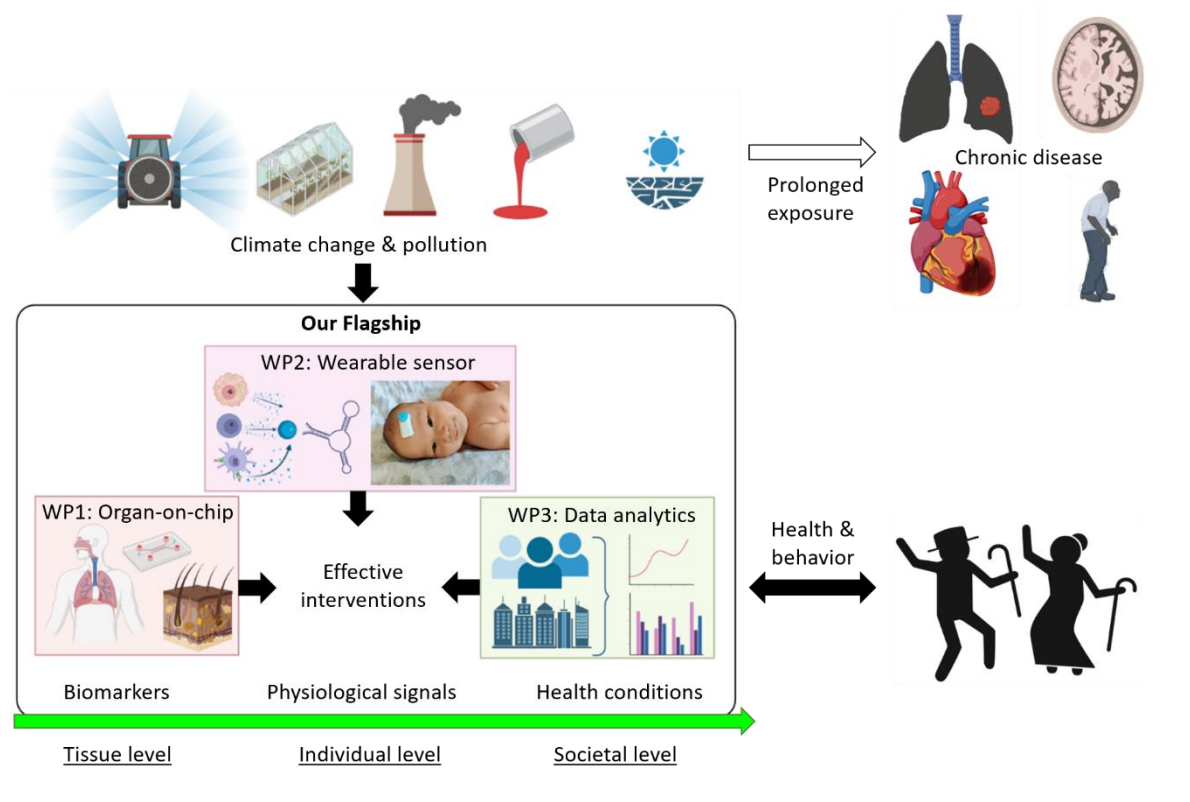
Prolonged exposure to excessive heat and environmental pollution is associated with increased risk of chronic diseases including cancer, cardiovascular, and neurological diseases. The current course of global climate change and decreased environmental quality puts 3.5 billion people in danger, disproportionately affecting disadvantaged populations and increasing health inequalities.

To significantly improve population health, targeted and timely interventions are key. Current research linking environmental stressors and health is, however, limited to long-after-the-fact large-scale statistics and long-term trends, restricting our ability to adapt behavior & provide cost-effective, timely, personalized prevention and treatment strategies.

In our Flagship, we propose a radically different approach (Scheme below). We advance and combine key developments in bioengineering, medical wearables and data-driven impact analysis – supported by clinical, economic and ethical insights – to **unveil and mitigate the effects of environmental stress on an individual scale in real time** from the tissue to the societal level:

1. Biological response at the tissue level: develop organ-on-chip platforms for in-depth investigations of the personalized biological response towards environmental stress.
2. Real-time response at the individual level: develop wearable skin patches for a) personalized, real-time biomarker sensing and b) monitoring and understanding individual behavioral response towards environmental exposure.
3. Behavioral and health response at the individual and societal level: combine personalized wearable sensors with advanced data analytics to develop individualized interventions and cost-effective policies.

These breakthroughs will **enable evidence-based personalized interventions**, providing citizens and clinicians with the toolsets and knowledge to implement personalized prevention and treatment strategies, while empowering policy-making agencies with analytics based on real-time data to **develop cost-effective and ethical policies for mitigating environmental health challenges**.



1.c Public Summary (max 300 words)

Our environment and our health are tightly interlinked. The alarming levels of climate change (heat) and pollution put about half the human population, 3.5 billion people, at increased risk for developing chronic diseases including cancer, cardiovascular, neurological, and autoimmune diseases. Now, more than ever, it is important for us to understand how our environment affects our behavior and health, and how we can mitigate its negative impact to live a healthy, long life.

Key to both effective medical intervention at the personal level, and effective policy making at the societal level, is that we can monitor the environment and its health effects **for many individuals in real-time**. To achieve this, our project will combine and advance three key technologies: organ-on-chip devices to understand the impact of environmental stressors, advanced wearable skin patches for continuous monitoring of a person's immune response and health behavior, and advanced algorithms to translate environmental monitoring and health monitoring into effective personal and societal interventions.

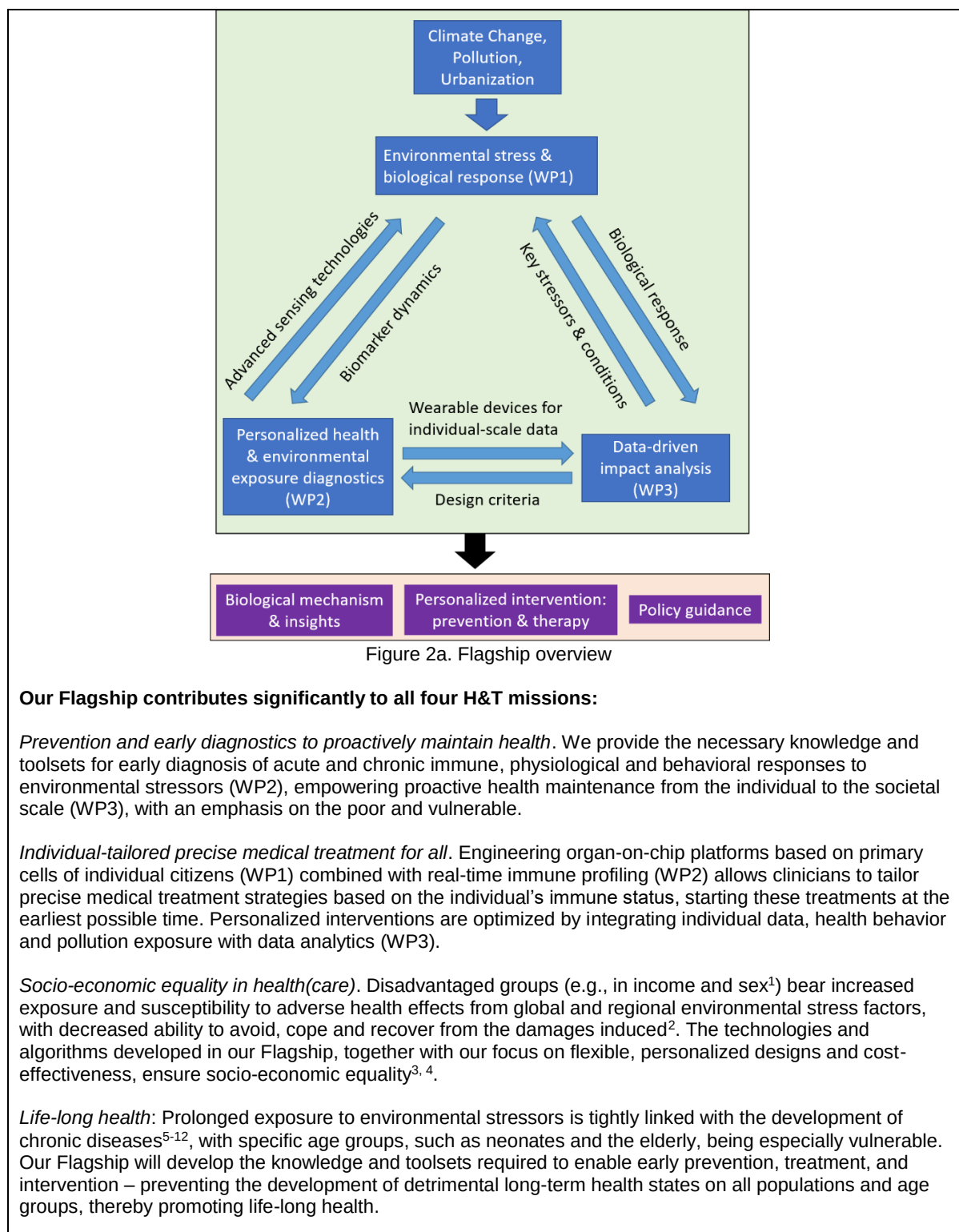
2. Relevance to Convergence Health & Technology mission (50%) – max. 2500 words for 2.a to 2.e

- *The Flagship program contributes to the Convergence Health & Technology mission (1)*
- *The Flagship program objectives fit with the guidance per theme provided by the theme leads, and/or fit with more than one theme of Convergence Health & Technology (2)*
- *The Flagship program merges approaches and insights from distinct disciplines to reach its goals (3)*
- *The staff granted by this call actively brings together different disciplines within the program, and creates visibility and attracts talents, partners, and funding for the program (4)*
- *The Flagship program has a clear view on long term sustainability after the 5-year central funding period (5)*
- *The envisioned results of the Flagship program have the potential for a significant contribution to healthcare, society and/or have scientific impact which has the potential for long-term societal impact (6)*
- *The Flagship program provides opportunities for BSc/BA and/or MSc/MA students of the participating institutes to participate in (interdisciplinary) projects (7)*
- *The proposal defines the pathway of how this contribution can be realized. The pathway includes concrete research and/or impact pathway (societal and/or economical) valorization strategies (8)*

2.a Contribution to H&T mission and fit with H&T theme(s)**Focus**

Unveiling the impact of environmental degradation (e.g., climate change, pollution) and the corresponding stressors (e.g., particulate matter, nitrogen oxides, heat) on health and utilizing this knowledge to further develop evidence-based personalized and societal interventions (cost-effective medical treatments and public policies) requires a deeper understanding of the biological, behavioral and medical effects of environmental stress, as well as the development of novel devices and algorithms – i.e. “Human-centered technology and AI for Health” at its finest.

Our Flagship enables new possibilities for personalized prevention, diagnostics, and care by developing effective testing platforms (i.e., organ-on-chip based on primary cells of individual citizens, WP1), novel medical devices (i.e., wearable sensors for real-time individualized immune profiling, WP2), and the necessary knowledge on cost-effectiveness of (personalized) interventions based on personalized data analytics on pollution, climate, and health (behaviors) (WP3) (Figure 2a).



2.b Synergy between disciplines

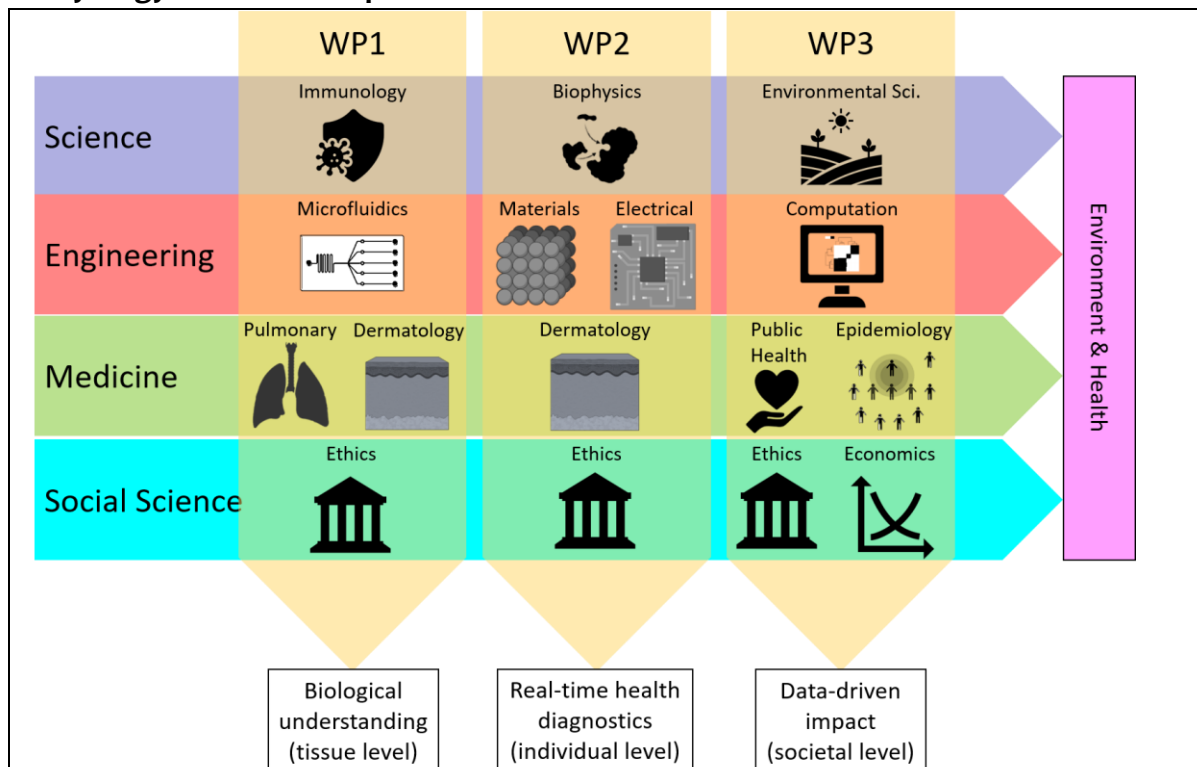


Figure 2b. Convergence of disciplines in our Flagship.

Figure 2b shows the interrelation in our Flagship between furthering biological understanding, real-time health diagnostics and data-driven impact. Below, we show that each of these themes requires the synergy of scientific, engineering, medical, and social scientific experts.

Biological understanding

Insights from **immunology** are critical: environmental stress activates immune and structural (e.g., fibroblast) cells to produce signaling molecules (including cytokines and secreted cell surface molecules) that collectively coordinate inflammatory responses¹³. Without timely intervention, this immunotoxicity may contribute to the development of many detrimental chronic diseases including cancer¹¹.

At the organ level, the skin and lungs are the first organs exposed to environmental stress^{8, 9}, with the skin as an easily and rapidly accessible organ for non-invasive health monitoring. Insights in **dermatology** and **pulmonary medicine** are therefore essential to further our understanding of human exposure to, and its corresponding response towards, climate change and pollution. Organ-on-chip systems are critical tools to enable fundamental biological insights under controlled parameters. This requires a synergy of expertise in **engineering** (e.g., nanofabrication, fluid mechanics) for fabrication of the system, **biology** (e.g., molecular genetics) for biological analyses, and physiology (dermatology and pulmonary medicine) for **clinical** insights.

Real-time health diagnostics

Novel biosensors that enable real-time health, behavioral and environmental monitoring are key to unlock the development of personalized interventions and effective policies. This requires **biophysics** to unveil the fundamental interactions between the target analyte (e.g., immunological biomarkers) and the bioreceptor (e.g., aptamers), as well as knowledge in **nanotechnology** and **materials science** to develop the sensing platform (e.g., for real-time, personalized mapping of immunological biomarkers). **Electrical engineering** expertise is prerequisite to embody the sensor into a wearable, wireless format with intuitive user interfaces. **Dermatology** is key to develop skin safety and clinical feasibility profiles of such devices, and **immunology** essential in generating physiological insights of the patient's real-time physiological state.

Data-driven impact based on individual-scale data

To develop evidence-based and (cost-)effective personalized and societal interventions – considering individual health and health behaviors in real-life conditions – requires expertise from a range of biomedical fields (i.e., **neonatology**, **pediatrics**, **pulmonary medicine**, **immunology**) as well as **econometrics**, **health economics**, **behavioral economics**, **urban economics** and **public health**. Underlying these analyses is the extension of the existing fixed environmental sensor network to include mobile sensors of pollution and heat and the development of new algorithms to obtain personalized pollution information combining our expertise from **data science** and **urban environment**. We include experts on **health economics** and **public health** to

identify and cater to vulnerable groups (e.g., neonates, elderly, but also certain ethnic or socioeconomic groups or deprived areas).

Our evidence-based approach yields personalized cost-effective interventions by combining public health knowledge with impact evaluation methods and measurement of preferences in the general population. Ethical integrity of our Flagship is safeguarded by including experts in **philosophy and ethics**.

2.c Long term sustainability plan

Overall feasibility

Our long-term vision is to establish a leading research and education center that attracts excellent national and international talent to contribute to our theme of 'Environment and Personalized Health' and our mission of 'establishing effective personalized interventions to prevent the negative health effects of environmental stress.' We have gathered a young and strong team that uniquely possess the scientific, technological, clinical, and philosophical expertise required to achieve these goals.

Funding perspective:

Awareness of the link between climate change, urbanization, pollution, and health (especially on chronic diseases) is rising and our key strategic missions align well with the current national and international research agenda (see Section 3d). Our flagship staff has demonstrated experience in attracting both industrial co-financing and publicly funded grants, ensuring that 1) five-year seed funding will be matched and 2) the foundations laid in these first five years will develop into an integrated research program.

Our flagship is not business-as-usual but a game changer in public health. With our general visibility measures as a catalyst (see section 2d), we will actively participate in discussions on national/international research roadmaps. Our aim, in five years' time, is that real-time personalized health monitoring to mitigate the individual and societal impact of climate change and pollution:

- is a specific mission of the HealthHolland Knowledge and Innovation Agenda (kennis- en innovatieagenda Gezondheid en Zorg)
- is well en-route to become a specific area of intervention in the Health Cluster of Horizon Europe.

People perspective:

Given our radical approach towards a long-standing and long-continuing health risk, we believe the Flagship program to be an excellent vehicle to put young(er) researchers in the spotlight while also building on established expertise within Convergence. Our balanced mix of tenured and tenure-tracked academics will ensure long-term continuity in people and fresh perspectives. Anchoring our Flagship into existing teaching programs of the three institutes (see Section 2d), we will train a generation of students and junior researchers with interdisciplinary skillsets that will impact the scientific, industrial, medical and general community.

Risk assessment

Risk	Mitigation
Delayed work package integration	- All workpackages contain tasks independent of other workpackages, ensuring basic output. - Establish contingency plans for inter-WP tasks.
Difficulty in obtaining the required external funding to support our long-term research activities	- Pro-active lobbying to have our unique individualized and real-time approach define the funding landscape. - Establish a Research Committee and an Outreach Committee (see Section 4d). - Engage industrial partners from project start. - Organize quarterly brainstorming sessions, thereby creating opportunities for major collaborative grants and a strong support network to secure funding. - Regular engagement with funding advisors of all three institutions. - Allocation of Convergence seed funding to refine grant proposal applications
Lack of visibility and impact	- Establish Committees for separate target groups: 1) Research Committee for academic impact 2) Education Committee for student impact; 3) Outreach committee for societal, industrial and governmental impact. - Yearly evaluation and brainstorm of strategies for visibility and impact.

2.d Visibility, attracting talents and new partners

Visibility strategy:

The main objective of our visibility strategy is to ensure that the project's outputs are disseminated to appropriate audiences to attract talents, partners, and funding. These activities also support our setting of the research agenda (see 2c)

Target Group 1: researchers in all relevant engineering, scientific, humanities, economics and medical disciplines, including chronic disease and environmental science specialists, to expand the scope of this Flagship to reach our long-term vision

Channels: open access scientific publications, conference presentations

Expected outcome/indicators:

- 30 scientific papers (6 per year) in leading journals, e.g., *Nature Biomedical Engineering*, *Journal of Allergy & Clinical Immunology*, *Journal of Human Resources*, *American Economic Journal: Applied Economics*
- 50 conference presentations (10 per year) at e.g., *European Public Health Conference*, *Materials Research Society Meeting*, *American Institute of Chemical Engineer Annual Meeting*, *European Academy of Allergy and Clinical Immunology Annual Congress*

Target Group 2: industry and general public

Channels: workshops with industry, open access scientific publications, conference presentations, science projects with primary/ secondary school students, student association, popular media

Expected outcome/ indicator (in addition to the above outcomes):

- Annual workshops with industrial partners
- Science projects with primary and secondary school students (e.g., The Globe)
- Engage with student associations (e.g., Geneeskundestudenten In De Samenleving and Green Team in EMC) for dissemination and translation of our work to society
- Project website, blog and social media, including youtube
- Press releases

Target Group 3: Policy advisors/governmental organizations

Channels: open access scientific publication, conference presentations, whitepapers

Expected outcome/indicator (in addition to the above outcomes):

- Workshops and roundtables to connect academics, policy makers, and industrial parties
 - Capitalize on existing consortia lobbying governmental organizations (e.g., Health Holland: human Organ and Disease Model Technologies (hDMT))
- Annual interactive meetings and brainstorming sessions with public health agencies and municipality representatives
- 2 whitepapers of our results for public health agencies

Education strategy:

We engage BSc/MSc students through:

- (1) Theses supervision. At least 10 MSc/BSc students from different disciplines within the three institutions will start every year. We will foster cross-institutional interdisciplinary interactions through joint supervision and progress meetings, monthly social events, and a journal club.
- (2) By year 3 (2025) we will kick off a joint minor at the bachelor level and an elective course at the Master level on Environment and Health.
- (3) Inter-institutional guest lecture, e.g., Biomaterials (TUD MSc course) and Biomedical Electronics (TUD MSc course), the public health and smarter choices for better health minor programs (EUR), economics of sustainability master's program (EUR), and health economics master's program (EUR).
- (4) Inter-institutional community projects for students.

And PhD/postdocs through:

- (5) A cross-institutional PhD course on 'Environment, Health, and Interventions'
- (6) Co-supervision by faculty members from at least two convergence institutions
- (7) Access to mentorship and resources throughout our Flagship
- (8) Monthly 'Environmental Health Impact' multidisciplinary seminar series alternating between participating institutions

2.e Long term societal impact

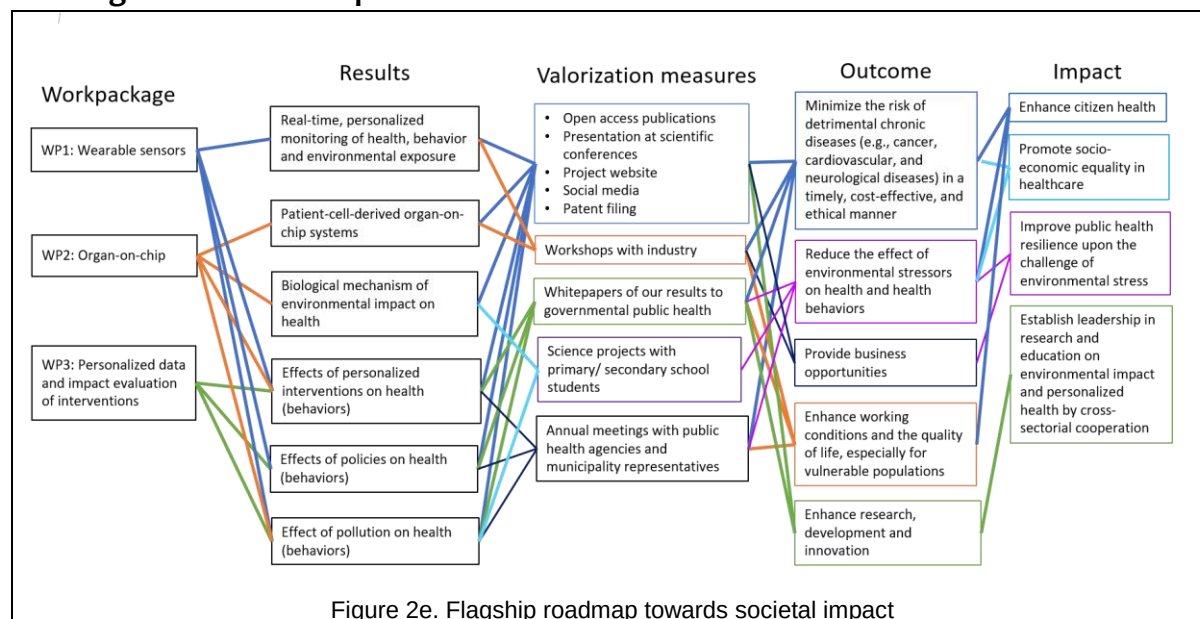


Figure 2e. Flagship roadmap towards societal impact

Figure 2e presents the detailed valorization roadmap on the following societal impact:

Enhance citizen health: the World Health Organization estimated that 1 out of 4 global deaths are related to environmental factors, with over 4 million due to air pollution alone¹⁴. Air pollution also accounts for the loss of 100+ million disability-adjusted-life-years¹⁵. The World Bank estimates that 6.4 million people died prematurely in 2019 due to exposure to air pollution. The health (mortality and morbidity) costs were equivalent to 6.9 percent of global GDP worldwide, or 2.8 percent in high income OECD countries¹⁶. Our Flagship aims to significantly reduce, and eventually relieve, our citizens from this economic, societal, and personal burden induced by environmental degradation.

Promote socio-economic equality: The health impacts of environmental degradation are often linked to environmental racism¹⁷ (e.g., race and ethnicity) and social determinants of health¹⁸ (e.g., education, income, and vocation). Environment-related health hazards also affect vulnerable populations more dramatically than privileged residents who have the money, resources, or political power to relocate, access better health care, and purchase higher quality food. Our Flagship aims to overcome these challenges by analyzing the environmental effects on health with a special emphasis on disadvantaged populations, thereby providing inclusive medical interventions on the individual level and holistic policy guidance on the societal level, promoting socio-economic equality.

Improve public health resilience: Our Flagship will develop technological toolsets, evidence-based knowledge to not only intervene, but also build preparedness of our society on the health effects of environmental degradation. The knowledge from our Flagship will also inform policies to prevent health deterioration due to pollution and climate change through: (1) personalized interventions on health (behaviors); (2) environmental and urban policies.

Establish leadership in research and education: with our unique approach in focusing on personalized, evidence-based interventions, along with our effective collection of multi-disciplinary talents, our goal is to establish the Netherlands as global leader in valuing and countering the health impact of the environment (e.g., climate change and pollution).

Valorization strategy

WP1: Industrial prototyping (Y5-Y7), followed by commercialization (Y7-Y10) of patented organ-on-chip systems.

WP2: Patent filing and engagement with the medical device industry for marketing, production, and commercialization in 10 years.

WP3: We will establish partnerships with public health agencies over the upcoming 10 years to provide effective intervention and policy strategies, promoting citizen health by combatting the effects of climate change and pollution.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

- The Flagship program contributes to excellent Convergence Health & Technology research, at the forefront of international research (1)
- The Flagship program is both innovative as well as challenging in its research and approach (2)
- The Flagship program has a clear strategy to engage students in its research plan (3)
- The Flagship program has a clear aim and long-term work plan, including, if applicable, a specification of required facilities and a data-and infrastructure plan. Coherence within the program is clearly demonstrated (4)
- The Flagship program aligns with or builds on (inter)national research agendas in this field, fit with the (inter)national research agendas but has a “high risk, high gain” main objective that is thoroughly explained (5)
- The Flagship program is feasible in terms of research performed, staff and budget needed (6)
- The Flagship program has a concrete plan for attracting external funding, which should result in at least the same amount as the central funding over the 5-year period (7)

3.a Ambitions

The Challenge: Environmental degradation generates stressors (e.g., pollutants, heat) that negatively impact human health¹⁹: persistent exposure increases the risk of chronic diseases including cancer¹¹, cardiovascular⁵, neurological²⁰, and autoimmune¹² diseases. Targeted and timely interventions may significantly improve survival and quality of life²¹. Current research on the links between health and environmental stressors are, however, limited to long-after-the-fact large-scale statistics²² and long-term trends²³, restricting the ability of clinicians to provide effective, timely, personalized prevention and treatment strategies, while disproportionally disadvantaging populations that are subjected to racism, poverty, sexism, and heterosexism².

Our Mission: Our Flagship aims to unveil and relieve the effects of environmental stress on health, especially for disadvantaged populations, using a radically new approach. We focus on the necessary technological, clinical, and scientific developments – supported by economic and ethical insights – to **unveil the effects of environmental stress on an individual scale in real time**. These breakthroughs will **enable evidence-based personalized interventions**, providing clinicians and public health agencies with the toolsets and knowledge to enforce cost-effective individually-tailored, timely prevention and treatment strategies to maintain health, while empowering policy-making agencies with analytics based on real-time data to **develop effective and ethical policies** sensitive to the health challenges of all citizens, including marginalized groups.

3.b Aims and feasibility

Biological understanding (WP1)

Challenge: The biological response upon exposure to environmental stress at the tissue level is unknown, hindering the development of mechanistic insights and intervention strategies.

Solution: Organ-on-chip platforms enabling in-depth investigations of the biological response towards environmental stress.

0 – 5 years: Focus on the development of skin- and lung-on-a-chip systems as these are the main organs interfacing with the environment.

5 – 10 years: Expand to a wider range of organ-on-a-chip systems (e.g., nose-, cornea-, gut-on-a-chip) and integrate the expertise of biologists, immunologists, and environmentalists to develop holistic and integrated knowledge on the impact of environmental stress and health.

Sensing technologies (WP2)

Challenge: Decode the health & behavioral response towards environmental stress at the individual level in a timely, non-invasive manner.

Solution: Wearable sensors for real-time monitoring of personal immune and physiological response, individual behavior and environmental signals.

0 – 5 years: Develop personalized skin patch for quantifying immunological biomarkers, providing real-time individual immune profiles.

5 – 10 years: Expand sensor function to include key physiological signals (e.g., heart rate, blood oxygenation), behavioral signals (e.g., energy expenditure) and environmental signals (e.g., UV and atmospheric components). The feasibility is high, given our track records in wearable devices^{24–26}.

10+ years: Include treatment capabilities (e.g., electrical stimulation) to mitigate negative immunological responses²⁷, providing a closed-loop immuno-modulating system.

Data analytics and interventions (WP3)

Challenge: Measure individualized health impact and determine evidence-based personalized interventions

Solution: Advanced data analytics

0 – 5 years: Establish causal evidence and relationships between individualized health, health behaviors and the environment utilizing existing medical and environmental data in combination with sophisticated mobile sensors. Develop experiments to measure citizens' preferences and cost-effectiveness of personalized interventions.

5 – 10 years: Establish design guidelines for further sensor development. Extend assessment of personalized interventions using developed sensors and additional (vulnerable) population groups. Advanced data analytics and algorithm development will furthermore help shape effective policies and personalized intervention strategies.

Health impact focus

0 to 5 years:

- Experts in dermatology, immunology, pulmonary medicine, neonatology, and pediatrics, with established networks – including medical experts, patient organizations, public health agencies, and medical companies.
- Focus on analysis and intervention development for air pollution and heat impact

5+ years expansion:

- Chronic disease specialists, including oncologists, cardiologists, neurologists, and gastroenterologists.
- Expertise for all kinds of pollution, including soil experts, hydrologists, and microbiologists

3.c Work plan

WP1-WP3 are technical workpackages, WP4 and WP5 coordination workpackages.

WP1: Organ-on-chip models for assessing biological response at the tissue level

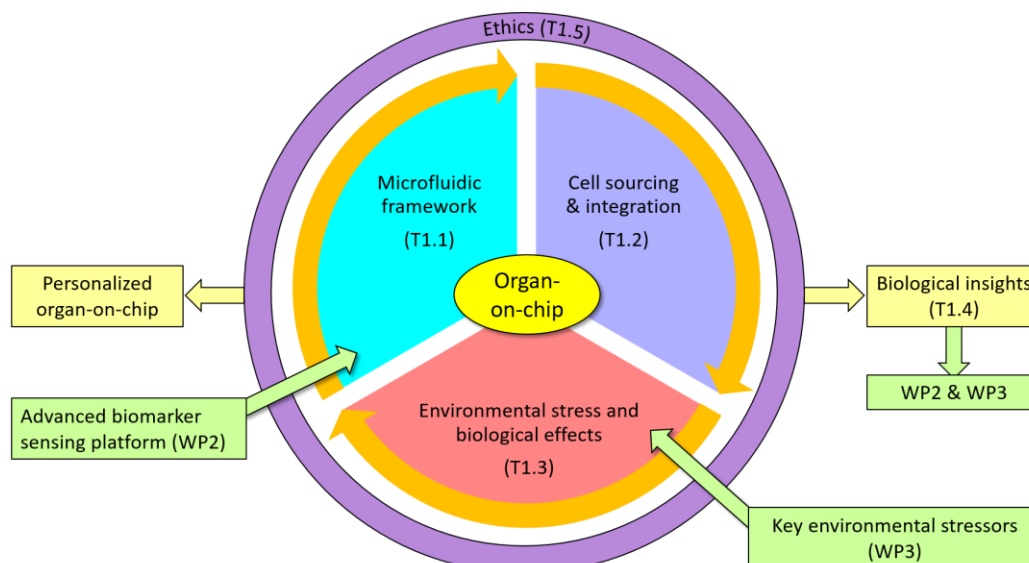
Skin- and lung-on-a-chip systems are currently TRL3 and expected to move to TRL5 by Y5 upon completion of system fabrication. Industrial engagement, especially with the pharmaceutical industry, is expected starting Y5 for animal-free, cost-effective, physiologically relevant platform of organ-on-a-chip systems.

Aims:

- Develop lung- and skin-on-chip models
- In-line analysis of culture medium composition (including cytokine concentration) and targeted proteomics on extracted tissues

Tasks:

- T1.1. Develop organ-on-chip microfluidic devices for skin and lung
 T1.2. Cell sourcing, culturing, and monitoring (e.g., with primary cells and induced pluripotent stem cells)
 T1.3. Quantify biological effects of environmental stressors with organs-on-chip
 T1.4. Translate organ-on-chip systems to WP3 and WP2 analyses
 T1.5. Environmental and social ethics of personalized organ-on-chip systems



WP1 work plan.

WP2: Wearable sensors for real-time analysis of environmental impact on individual health

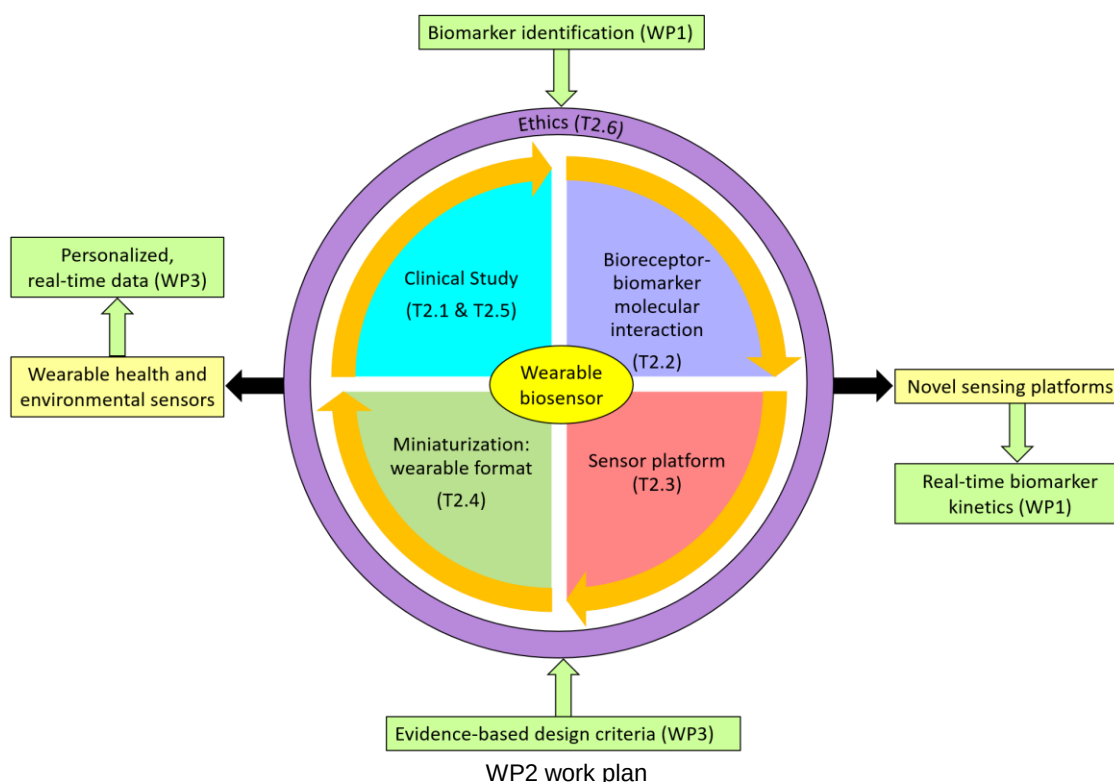
We will initially focus on cytokine aptasensors that are currently TRL2, moving this up to TRL5 by Y5. This will then be further combined with other sophisticated sensors on the TRL7-level (e.g., for heart rate, blood oxygenation, and motion).

Aims:

- Understand levels of immunological biomarkers in non-invasively attainable biofluids (e.g., sweat) for normal and acute inflammation conditions
- Build a non-invasive, skin-safe biosensor capable of detecting and quantifying specific immunological biomarkers

Tasks:

- T2.1. Assess biomarker concentration in sweat on human subjects
 T2.2. Investigate the change in structure of bioreceptors upon contact with analyte based on single-molecule fluorescence resonance energy transfer²⁸
 T2.3. Develop optical and/or electrical detection method for analyte detection
 T2.4. Engineer the sensing platform into a wearable device
 T2.5. Clinical testing of wearable device for skin safety & feasibility
 T2.6. Environmental and social ethics of wearable sensors

**WP3: From the lab to the population. Personalized data and impact evaluation of interventions**

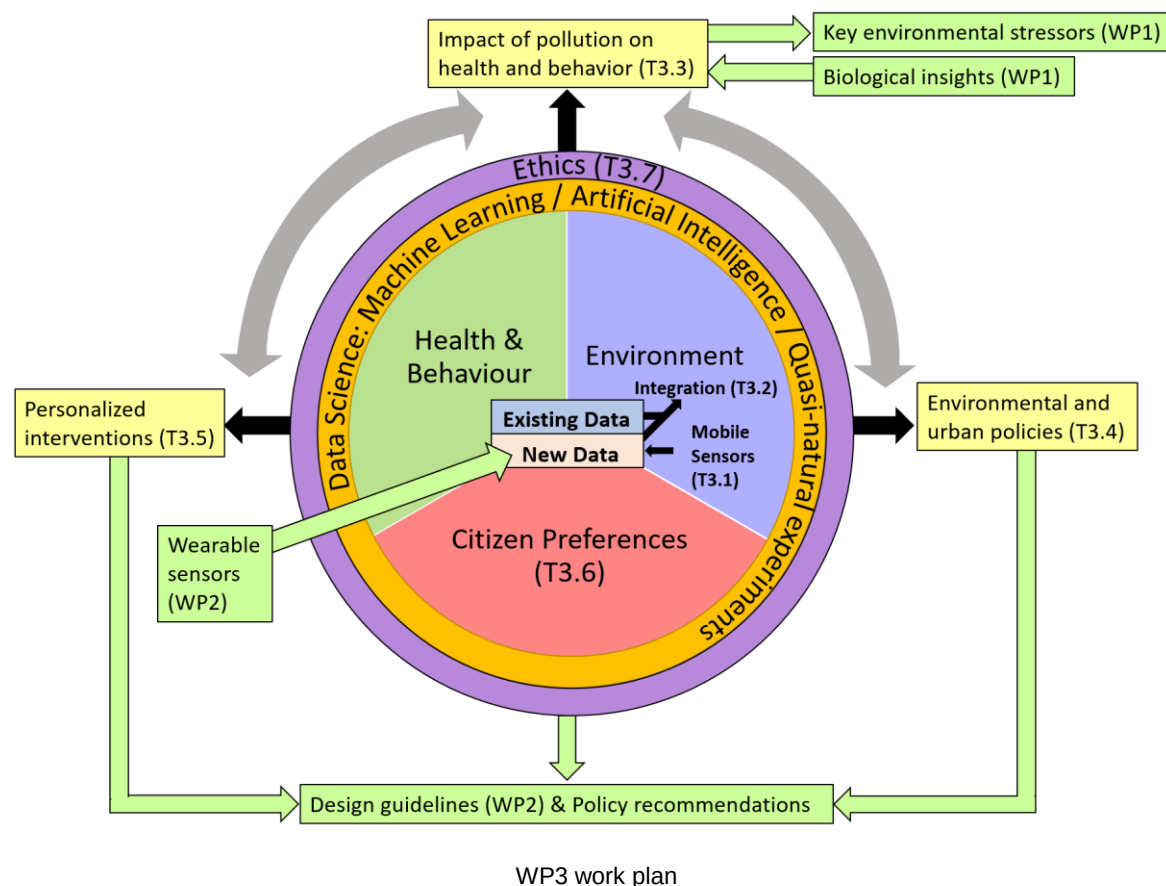
We will provide evidence to facilitate effective TRL progression in WP1 and WP2, informing key players (e.g., policy makers) on the importance of personalized interventions for environmental impact mitigation.

Aims:

- Improve data, strategy and methods to obtain more localized/personalized information and forecasting on pollution and climate (e.g., from mobile sensors)
- Obtain causal evidence on the effect of policies to reduce pollution and personalized interventions on health and health behaviors
- Distinguish between immediate and long-term effects on health and health behaviors
- Investigate preferences for health behaviors under different climate circumstances and for adopting wearable technologies that inform about such circumstances
- Investigate preferences for evidence-based policies to mitigate (the impact of) climate change and pollution on health and health behavior

Tasks:

- T3.1. Extend fixed sensor network with existing air quality mobile sensors
 T3.2. Include mobile sensor data with gridded model data²⁹ using machine learning and artificial intelligence³⁰
 T3.3. Evaluate effects of pollution on health in both the Netherlands and other countries, including low- and middle- income countries
 T3.4. Estimate impact of policy interventions on health and health behaviors
 T3.5. Estimate impact of personalized interventions on health and health behaviors for various populations
 T3.6. Quantify public preferences regarding i) uptake of the new technology developed in WP2 to assess its marketability; ii) active behavior in response to outdoor circumstances; iii) preferences for different policies, considering their costs and benefits
 T3.7. Ethical guidance on policy design

**WP4: Dissemination and public outreach**

Aim: Establish external collaboration opportunities and ensure communication of project outcomes to a wider audience.

Tasks:

- T4.1. Project website and social media
 T4.2. Intellectual property strategy and management
 T4.3. Workshops for industry
 T4.4. Workshops for early-career researchers
 T4.5. Seminars on environment and health
 T4.6. Whitepapers for governmental public health agencies

WP5: Project management

Aim: Ensure effective Flagship management and sustainable Convergence

Tasks:

- T5.1. Data management plan
 T5.2. Internal project communication and meeting organization
 T5.3. Implement and maintain internal and external collaborations
 T5.4. Financial management
 T5.5. Archiving results
 T5.6. Time management of deliverables and reporting
 T5.7. Overall administration

Gantt chart

Task	Y1	Y2	Y3	Y4	Y5
T1.1.					
T1.2.					
T1.3.					
T1.4.					
T1.5.					
T1.6.					
T2.1.					
T2.2.					
T2.3.					
T2.4.					
T2.5.					
T3.1.					
T3.2.					
T3.3.					
T3.4.					
T3.5.					
T3.6.					
T3.7.					
T4.1. - T4.5.					
T4.6.					
T5.1.					
T5.2. - T5.7.					

Risk and Mitigation

Risk	Mitigation
Delayed project license for human clinical studies	Initiate license application process at project start, using our previous protocol as reference ²⁴ .
Failure to obtain valid physiological information with the developed wearable device	Ensure the wearable device is fully optimized in function and safety with comprehensive in-lab experiments and enforce fast feedback between the clinicians and the engineers for solutions
Insufficient amount of cytokines in sweat or not enough sweat production	Probe alternative body fluids (e.g., design microneedles for intradermal fluid extraction) and optimize the data processing algorithm to enhance sensitivity and accuracy.
Difficulty in reproducing relevant physiology on organ-on-chip	Focus on most important features of the tissues relevant to this Flagship (e.g, immunological response), simplifying other features if needed.

External Funding Plan

Present round funding will be matched by TNO (in-kind contribution of 25k per year for the duration of this seed-funding period) and existing public funding (Table 1). Flagship consortium members have a proven track record to acquire external funding (see 4a), with several co-funding opportunities (Table 2) still awaiting decision. We will furthermore solicit relevant companies and public agencies (see Section 4c).

The five-year seed period will be used to setup the overall program structure and hire a minimum of 10 PhDs and 5 post-docs. Longer-term funding efforts will be initiated on program start and serve to expand our Flagship program.

Table 1. Existing public funding:

Funding Agency	Program	Tasks	Contribution to Flagship	Researcher
NWO	Vici	T2.2	1 PhD	Joo
NWO	Open Competition, Domain Science XS	T3.2-T3.4	6-month researcher	Dedoussi
Horizon Europe	Green airports and ports as hubs for sustainable and smart mobility	T3.4	1 research assistant	Bornioli
NWO	NWA-ORC	T1.1	1 PhD	Mastrangeli

Table 2. Examples of future funding plan

Funding Agency	Program	Project	Amount	Tasks	Submission
NWO	NWA-ORC	Pregnancy outcomes during a pandemic response	1.1 M	T3.3-T3.4	Submitted
ESA	EO Science for Society	Anchoring Satellite Observations to Citizen Science Sensing	150k	T3.1-T3.2	Submitted
Horizon Europe	HORIZON-CL5-2022-D5-01-12	Silent and ultra-low local air pollution aircraft	260k	T3.2-T3.4	April 2022
NWO	ENW-XL	Aptasensor for immunological biomarker detection	3M	T1.2, T2.2-T2.4	2023
NWO	NWA-ORC	Organ-on-chip for immunological mechanistic studies	1M	T1.1-T1.4	2023
NWO	Open Technology Program	Wearable sensor for immune profiling	850k	T2.3-T2.4	2023

Student and early-career researcher involvement plan

- Professional development will be fostered through Flagship-wide educational workshops, journal clubs, and grant-writing workshop.
- Students and early-career researchers will collaborate through joint progress meetings, brainstorming sessions for project ideas and approaches, and our 'Environment and Health' seminar series.
- PhD-students will be co-advised by faculty members from at least two institutions.
- Multidisciplinary bi-monthly meetings allowing MSc students to share feedback and exchange ideas.
- Where possible, MSc-students will study related research questions to enhance this exchange.

3.d Alignment of the program with relevant national and international research agendas

Our Flagship fits with the following (inter)national research agendas:

- EU: The Environment and Health commission has established a strategic research and innovation agenda to “protect citizens' health from Impacts of environmental degradation”²¹.
- WHO: The 'Ostrava Declaration on Environment and Health' has stimulated governments across the globe to strengthen the 'adaptive capacity and resilience to health risks' related to environmental factors²².
- Dutch Ministry of Economic Affairs and Climate Policy: The 'long term strategy on climate mitigation' guides the Integrated Knowledge and Innovation Agenda (IKIA) for adaption and mitigation strategies towards the environmental stress induced by climate change.
- NWA: The 'Climate Adaptation and Health' program as well as the 'Prevention Knowledge Agenda', HealthHolland's 'Knowledge and Innovation Agenda for Health and Health Care'.
- ZonMw: 'Climate and Health' program with annual calls between 2019 and 2023.
- Chronic disease development tightly aligns with the missions of KWF (Dutch cancer society), Lung Foundation Netherlands, and the European Union (e.g., the Cancer Mission).

4. Consortium (25%) max 2000 words for 4.a to 4.d

- The Flagship program is driven and executed by a well-organized team of scientists from all three institutions and at least 4 departments or faculties in total. In exceptional cases, it is possible to form a Flagship program with only two leads from two institutions, but this should be clearly motivated (1)
- Members of the consortium have a track record of excellence in academic research, translation of scientific results into clinical application or societal impact (2)
- Members of the consortium have a significant track record in engaging and supervising BSc and MSc students in research projects (3)
- Members of the consortium have demonstrated to be able to attract external funding, thereby enhancing the programs sustainability after the 5 years of central funding (4)
- There is a demonstrable fit between the aims and objectives of the program and the expertise of the consortium members (5)
- End users (private and/or public sectors) are involved from an early start and team members are well-connected to external partners in the ecosystem (6)
- Governance should be clearly described and is organized in such a way that it facilitates collaboration and optimally makes use of the expertise of all participants (7)

4.a Track record of the consortium members

Our multidisciplinary convergence of the three Institutions is evidenced by the fact that all three Institutions have significant representation (each close to 1/3) in our Flagship Team (Figure 4a).

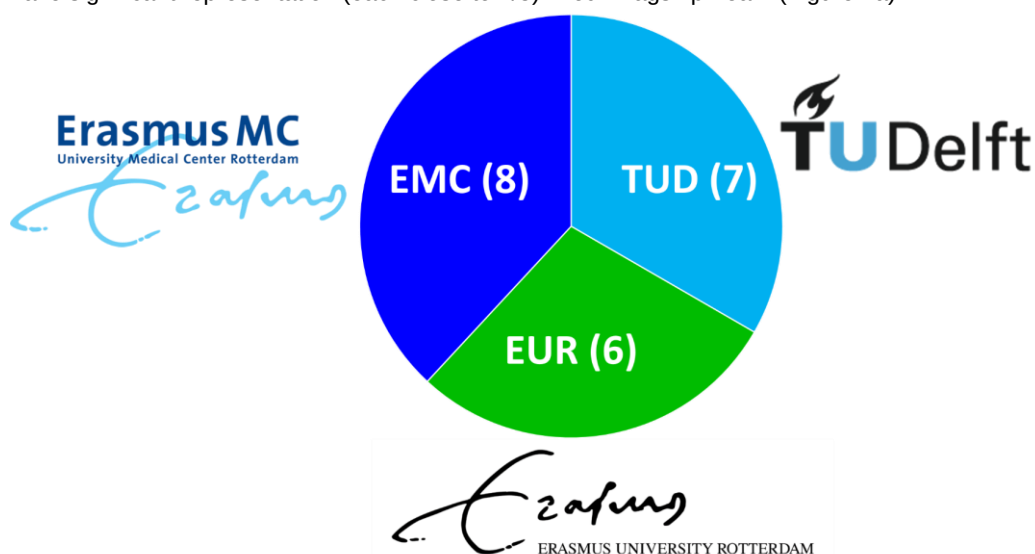


Figure 4a. Institutional convergence. The number of applicants from each Institute is represented in brackets.

Table 4.1 showcases our strong track record in academic achievements across multiple disciplines (e.g., published in high impact journals including *Nature*^{31, 32} and *Cell*³³), external funding attraction (a total of >54 M€ in the past 10 years), education (e.g., supervised >1250 PhD/MSc/BSc/medical students), and translation (e.g., co-founding of start-up companies³⁴, generation of patents³⁵⁻³⁷, national and regional advisor boards).

Last Name	Academic Profile	Translation into clinical application and social impact	Education (number of MSc/BSc students, and other education activities)	External Funding	External Collaborations
Been	Perinatal and pediatric public health ³⁸⁻⁴⁵	Worked with WHO on tobacco impact on child health ⁴⁶ Advisor to RIVM on ultrafine particles impact evaluation Chair Taskforce Smoke-free Erasmus MC ⁴⁷	10 PhD 27 MSc/BSc 3 community projects	~2.2 M€ ZonMw, CSO Scotland, International Pediatric Research Foundation, Trasher Fund, European Respiratory Society, KNAW, Longfonds, FNO	Care4Neo, Rotterdam Municipality
Bornioli	Sustainable urban/transport policies ⁴⁸⁻⁵⁰	Development and evaluation of sustainable transport and urban policies. Development and implementation of private companies' strategies for emission reduction.	10 MSc/BSc Lecturer at UK universities	~1 M€ H2020, funding from companies	Rotterdam Gemeente, Public Health England, Bristol City Council, BMW, Parkmobile
Burdorf	Environmental impact on population health ^{6, 7, 10, 39, 51, 52}	Chair of Netherlands Health Council committees 'Health benefits through cleaner air' (2018) and 'Risk of ultrafine particles in the outside air' (2021). Chair of Sustainable Erasmus MC, which is involved in Rotterdam 'climate table'	> 40 PhD	~1.3 M€ ZonMw, EU Joint Programme Initiative, EIT Health	City of Rotterdam, Zeeland Public Health Service, and Dienst centraal milieubeheer Rijnmond (DCMR)
Chaves	Health effects of circadian disturbance ⁵³⁻⁵⁵	Animal study into jet lag health risks	>25 BSc/MSc	~1.1 M€ NWA-ORC, NWO-ECHO, EUR Fellowship	UvA-SILS UMC Utrecht
Dedoussi	Human health impacts & societal cost of air pollution ^{32, 56, 57} Atmospheric impacts of aviation emissions ^{58, 59}	Contributions to: - UN's ICAO first global CO ₂ regulation for aviation - NO _x calculations Lelystad airport; National advisory board nitrogen problem aviation sector	4 PhD 12 MSc	~850 k€ H2020(2x) NWO: Veni, NWO-XS(2x)	n/a
Dik	Cellular and molecular aspects of inflammatory disease ⁶⁰⁻⁶²	Immunological diagnostic tests into patient care. Member national advisory board newborn immune disease screening program.	n/a	~1.3 M€ ZonMW-TZO, Uitzicht, International Retinal Research Foundation	n/a

		Raising public awareness on immune system risks			
Garcia-Gomez	Health Economics: applied microeconometrics ^{63, 64} , quasi-natural experiments ⁶⁵ ; health disparities ⁶⁶ .	Expert member Spanish government on 2050 agenda Collaboration with UWV on disability policies ⁶⁷	3 PhD >40 MSc/BSc Academic director Health Economics Masters	n/a	n/a
Gianoli	Urban environment, urban climate change, statistics, econometrics, machine learning	Involved in >20 multi-year advisory and research projects related to urban environment and urban climate change.	10 PhD >120 MSc MSc program coordinator	~2 M€ European Commission, World Bank, United Nations Environment Program	Various European and Asian academic institutes
Hermans	Mast-cell-related diseases, from bench to bedside ^{60, 68-70} .	Clinician in Allergy & Clinical Immunology, access to test populations.	8 BSc/MSc	Novartis MPN patient society (15 k€)	UMC Groningen
Joo	Single molecule techniques: CRISPR ²⁵ and Argonaute ²⁶ . First-ever high-throughput single-molecule fluorescence.	n/a	>35 MSc/BSc	~9 M€ ERC: Starting, Consolidator NWO: Vidi, Vici, Vrije Program (currently called ENW-XL, project leader)	n/a
Los	Climate change and air pollution effects, citizen science ⁷¹	Team leader air pollution forecasting / citizen science initiatives	>20 MSc MSc program coordinator	~1 M€ H2020, ESA, Dutch regional support	Waag Society Amsterdam, WUR, various municipalities (NED, GER)
Mastrangeli	Micro-electromechanical organs-on-chip ⁷² , capillary micromanipulation ⁷³ , soft robotics ⁷⁴ , nanofabrication ⁷⁵	n/a	8 PhD 40 MSc/BSc	~1.5 M€ NWA-ORC, TTW Perspectief, industrial project with Nexperia.	Nexperia, Philips, micronit, TNO, Multichannel Systems, Besi, Microfluidic ChipShop, BI/OND
Muratore	Microelectronics: (bio)sensor interfaces ^{76, 77}	Patent ³⁷ licensed by Blackrock Microsystems (clinical neural interfaces)	8 MSc	\$150k Wu-Tsai Neurosciences Institute	Infineon, Analog Devices

	Edge algorithms for implantable and wearable devices ^{37, 78, 79}				
Richie	Just and sustainable health care ^{1, 80-83}	Head of the Environmental Bioethics Affinity Group (ASBH) Joint-Editor <i>Global Bioethics</i> COP26 Green Zone participant	5 BSc	n/a	Erasmus Medical College Center for Bioethics and Human Dignity at Trinity International University (Deerfield, IL)
Rottier	Advanced human lung models ⁸⁴⁻⁹⁰	n/a	15 PhD >25 BSc/MSc	>4.5 M€ ZonMw, NWO, Longfonds, Proefdiervrij	LUMC, U-Maastricht, U-Twente
Rwei	Biomaterials: externally-triggerable drug delivery systems ⁹¹⁻⁹⁴ . Biosensors: bioelectronic skin patches for real-time, monitoring of physiological signals ^{24, 25} .	Clinical testing of wearable devices; a patent ⁹⁵ licensed to a startup (Sibel Health) ⁹⁵ . Co-founded an award-winning startup (Aqua fresco) ²³ with a granted patent ²⁴	10 MSc/BSc	\$100k NIH (USA) TL1	AstraZeneca, Merck KGaA
Stadhouders	Genetic susceptibility to inflammatory disorders ⁹⁶ . Gene expression and cell identity ^{97, 98} . Lymphocyte activity in inflammatory disease and cancer ^{99, 100} .	n/a	8 MSc/BSc	~3.34 M€ NWO VIDI, KWF, Dutch Lung Foundation	CRG Barcelona, Harbour Biomed
Staufer	Micro- and nano-engineering ^{101, 102} , microfluidics ¹⁰³ in bio/health ¹⁰⁴⁻¹⁰⁶ , physics ^{107, 108} , and planetology ¹⁰⁹⁻¹¹¹ applications.	Collaboration with clinicians for developing arthroscopic instruments and microneedle arrays.	20 PhD 40+ MSc	~5.5 M€ STW, NWO TTW, NanoNextNL, KTI (CH), SNF (CH), WN foundation	Nanosurf Nanoworld, Nanotools, SIOS, Flubetech, NTC Weitz MA3 Solutions, Tegema Delmes, Maypa, Maastricht Instruments, Bronkhorst, Micronit, MyLife Technology, VSL, IBM, Intel, NASA, JPL,

					ESA, 10 universities
Thio	Dermatology Role of gut and skin microbiota¹¹² and circadian clock in infectious¹¹³ and atopic¹¹⁴ diseases.	Informing general public and medical professionals through various channels.	10 PhD 40 BSc/MSc 200 dermatology interns 60 dermatology residents	~700 k€ Indonesia Endowment Fund for Education	AbbVie, Novartis, LEO Pharma, Eli Lilly, Boehringer Ingelheim, Janssen (J&J), UCB, Almirall, Pfizer
van der Horst	Chronobiology and health: mouse models, cultured tissues, transcriptomics, epigenetics, <i>in vivo</i> imaging¹¹⁵⁻¹¹⁸.	Creating public and professional awareness (e.g., top 5% best-viewed YouTube lecture¹¹⁹, radio, TV, newspapers¹²⁰). Member Netherlands Health Council committee 'Health risks of night shift work' Translation into personalized medicine, chronotherapy.	>50 BSc/MSc/PhD	~14 M€ NWO (various, including Vici), NWA-ORC, NGI, AICR.	DutchSail KRVE (boatmen Rotterdam harbor) Methylomics Propeaq
van Exel	Health economics, HTA, behavioral economics, choice modelling, participatory research / public engagement	Various expert committees and advisory boards for governmental, societal and private organizations. Publications in national policy (and professional) journals, translating academic findings to target groups). Guest lectures for societal organizations. Commercial projects / implementation / consultancy. Open science initiatives.	27 PhD 83 MSc/BSc Chair of the Board Master Health Economics and Management. MSc and BSc program and course coordination.	>5 M€ ZonMW, NWO, EU (Horizon, MCSA), European Commission, MRC (UK), Erasmus Trustfonds, several pharmaceutical companies	VU Amsterdam University of Amsterdam Radboud University >10 European universities RIVM SCP TNO EHRO iMTA European Patient Federation Dutch Patient Federation MantelzorgNL

					ECORYS
Van Ourti	Applied health Economics ^{121 122} , quasi-natural experiments ¹²³ , health Disparities ¹²⁴ , preventive care ¹²⁵ , perinatal health ¹²⁶	Management Team of EUR Smarter Choices for Better Health Initiative Evaluator for Belgian Health Care Knowledge Centre	10 PhD >75 MSc/BSc	n/a	n/a

4.b Expertise of the consortium members in relation to the aims and objectives of the program

Our consortium will utilize an integrated strategy to decode and mitigate the environmental impact on human health on three levels:

- (1) **Biological response at the tissue level.** Mastrangeli (TUD, micro/nanofabrication, micromanipulation, soft robotics) and Thio (EMC, dermatology) will lead the development of skin-on-chip devices. Staufer (TUD, micro and nanofabrication; microsystem design; microfluidics), Stadhouders (EMC, epigenetics), and Rottier (EMC, developmental and molecular biology of the lung, primary lung cells, induced pluripotent stem cells) will lead the development of lung-on-chip devices. The immunological analysis of biological response upon exposure of the developed organ-on-chip systems to environmental stress will be evaluated by Hermans (EMC, clinical immunology). Chaves (EMC, molecular and cell biology) will evaluate the molecular and genetic impact. Van der Horst (EMC, molecular biology and chronobiology) will serve as advisor on the decoding of biological mechanisms.
- (2) **Physiological response at the individual level.** Joo (TUD, single molecule technology) will provide fundamental insights on the interaction between the bioreceptor and the target immunological biomarker, with which Rwei (TUD, biosensor and biomaterials) will develop the optical/electrochemical sensing platform. Muratore (TUD, electronic engineering) will miniaturize this sensing platform into a wearable skin patch, the safety and clinical feasibility of which will be evaluated by Thio (EMC, dermatology) and Dik (EMC, clinical immunology).
- (3) **Behavioral and health response at the individual and societal level.** Los (EUR, air pollution/climate modelling) and Gianoli (EUR, statistics, econometrics, and machine learning) will integrate local and mobile sensor data into air pollution models and extend model capabilities using statistical methods to produce personalized air pollution information. Dedoussi (TUD, atmospheric impacts and aviation emissions) will contribute to the air quality measurement and modelling developments. Bornioli (EUR, health evaluation of air pollution interventions) will work on implementing mobile sensors, evaluating the effect of policies, and assessing consumer preferences on interventions. Van Exel (EUR, choice experiments) will explore the uptake of the developed sensor technology and its potential impact on health behavior, and use participatory evaluation techniques to study citizens' preferences for policies to reduce (the impact of) climate change and pollution on health and health behavior. Garcia-Gomez (EUR, applied microeconometrics with large linked datasets) and van Ourti (EUR, quasi-natural experiments) will coordinate the linkage of personalized pollution and health data and estimate the causal effect of pollution on health and health behaviors and the policies to reduce (the impact) of climate change and pollution on health and health behaviors. They will work in close collaboration with Been (EMC, perinatal and paediatric public health) and Burdorf (EMC, environmental impact on population health) to define the appropriate populations and outcomes. Finally, Richie (TUD, ethics and philosophy) will contribute ethical analysis of the technology, policies, and intervention strategies developed in this Flagship.

4.c Involvement of partners

TNO will be involved with T3.1 and T3.2 from project start. They will offer model data and support for the co-supervision of a PhD student.

We will also reach out to the following relevant companies with the indicated timeline based on existing connections of team members (if applicable):

- Merck KGaA (Y1, Rwei)
- Bayer (Y1, Thio)
- Dermavant Sciences (Y1, Thio)
- KLM (Y1, Thio)
- KPN (Y1, Bornioli)
- SomaLogic (Y2, Joo)
- NXP Semiconductors (Y2, Muratore)
- Demcon (Y2, Hermans)

- Organ-on-chip companies (Y2, outreach required)
- BioNTech (Y3, outreach required)
- Nostics (Y3, outreach required)

We will engage the following public parties:

- RIVM, DCMR (Y1, Van Exel and Burdorf)
- Municipalities (Y1, Burdorf)
- Waag (Y1, outreach required)

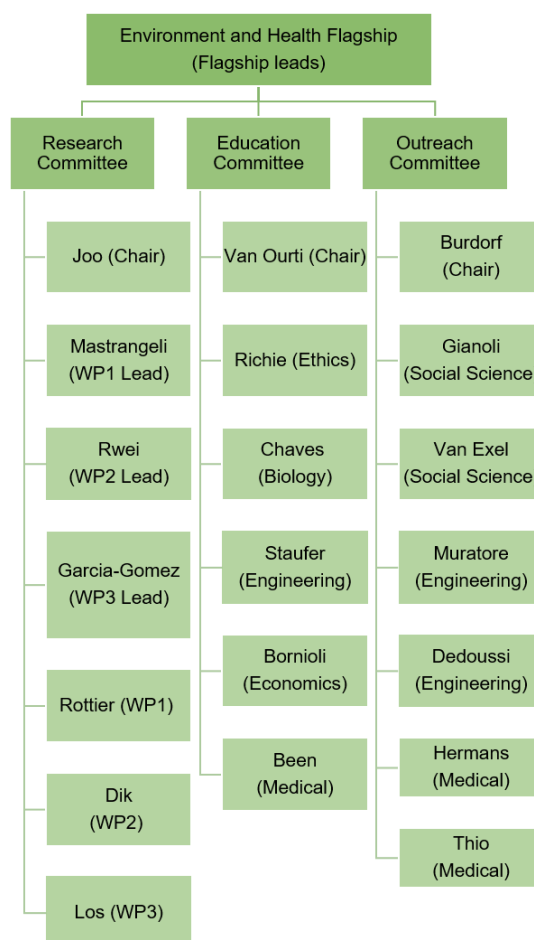
4.d Governance, organization and collaboration strategy of the consortium

The Flagship will be composed of three committees, overseeing our research, education, and outreach activities (see Figure below).

The Research Committee oversees the Flagship's research progress, fosters internal and external academic collaboration opportunities, and maintains close contact with the funding managers of each Convergence institution to maximize scientific impact and public funding attraction. Its chair, Joo (TUD), has a strong record in research (including publications in *Nature*³¹ and *Cell*³³) and attracting external funding (e.g., Vidi/Vici, ERC Starter/Consolidator, ENW-XL).

The Education Committee oversees the execution of the Flagship's education strategy, as per Section 2d. Its chair, van Ourti (EUR), is experienced in organizing PhD summer schools and he has a significant track record in supervising BSc/MSc students. He is experienced in mentoring early-career students in multidisciplinary research from his leadership role in the EUR Smarter Choices for Better Health Initiative.

The Outreach Committee will oversee the execution of the industrial, general public, public health and governmental agency outreach strategy (see Section 2d). It will be chaired by Burdorf (EMC), who is well-connected to key external partners (e.g., DCMR¹⁰, Zeeland Public Health Service^{6, 7, 51, 52}, City of Rotterdam), and supported by committee members with extensive networks.



Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

II Letters of intent and letters of commitment

III Letter of support from department head of the main lead

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Budget formulier: Convergentie Flagship Call

Flagship:	Personalized, Real-Time Health Impact of Climate Change and Pollution
Naam Penvoerder:	Ains Ruel
Startdatum en looptijd van het project:	1-9-2022, 5 years
Totaal project budget:	€ 4.974.099

R&D Overview: kostensoorten per type onderzoek (hele project periode)

1.	Loonkosten:						2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner			
	Partner die de kosten maakt	Functie	Aantal fte	Startdatum	Einddatum	Inschaling	Maandbedrag	Kosten																
	Onderzoeksorganisaties																							
	TU Delft	PhD	1.0	1-9-2022	31-8-2026		€	4,653	€ 223,344											€ 0	€ 0	€ 0.00	Convergence	
	TU Delft	PhD	1.0	1-9-2022	31-8-2026		€	4,653	€ 223,344											€ 0	€ 0	€ 0.00	Convergence	
	TU Delft	PhD	1.0	1-9-2022	31-8-2026		€	4,653	€ 223,344											€ 0	€ 0	€ 0.00	Convergence	
	TU Delft	PhD	1.0	1-9-2022	31-8-2024		€	4,653	€ 111,672	€	18,612		€ 55,836	€ 37,224	€ 0					€ 111,672	€ 0	€ 111,672.00	existing NWO grant TUD-BN VICI Chirimin Joo	
	TU Delft	PhD	1.0	1-9-2022	31-8-2024		€	4,653	€ 111,672	€	18,612		€ 55,836	€ 37,224	€ 0					€ 111,672	€ 0	€ 111,672.00	existing NWO grant TUD NWA-ORC LymphChip	
	TU Delft	postdoc	1.0	1-9-2024	31-8-2027		€	6,313	€ 227,268				€ 25,252.00		€ 75,756		€ 75,756.00		€ 50,504.00	€ 0	€ 227,268	€ 227,268.00	external future funding	
	TU Delft	postdoc	1.0	1-9-2022	28-2-2023		€	6,313	€ 37,878	€	25,252		€ 12,626							€ 37,878	€ 0	€ 37,878.00	existing NWO grant TUD: NWO open competition, domain science XS	
	TU Delft	project manager	0.25	1-9-2022	31-8-2027		€	7,668	€ 115,020											€ 0	€ 0	€ 0.00	convergence	
	Erasmus MC	Clinical researcher	1.00	1-9-2022	31-8-2024	Schaal 10		€	4,802	€ 172,872										€ 0	€ 0	€ 0.00	Convergence	
	Erasmus MC	PhD	1.00	1-9-2022	31-8-2026	OIO		€	3,226	€ 232,278										€ 0	€ 0	€ 0.00	Convergence	
	Erasmus MC	PhD	1.00	1-9-2023	31-8-2027	OIO		€	3,226	€ 232,278	€	16,082.67		€ 51,318		€ 58,705.33		€ 62,590		€ 43,980.67	€ 232,277	€ 0	€ 232,277.00	external future funding
	Erasmus MC	postdoc	1.00	1-9-2024	31-8-2027	Schaal 12.5		€	6,407	€ 345,961			€ 36,834.00		€ 112,069		€ 116,888		€ 80,170		€ 0	€ 345,961	€ 345,961.00	external future funding
	Erasmus Universiteit Rotterdam	PhD	1.00	1-9-2022	31-8-2026		€	4,653	€ 223,344											€ 0	€ 0	€ 0.00	Convergence	
	Erasmus Universiteit Rotterdam	PhD	1.00	1-9-2023	31-8-2027		€	4,653	€ 223,344											€ 0	€ 0	€ 0.00	Convergence	
	Erasmus Universiteit Rotterdam	PhD	1.00	1-9-2022	31-8-2026		€	4,653	€ 223,344	€	18,612.00		€ 55,836		€ 55,836.00		€ 55,836		€ 37,224.00	€ 223,344	€ 0	€ 223,344.00	external future funding	
	Erasmus Universiteit Rotterdam	PhD	1.00	1-9-2023	31-8-2027		€	4,653	€ 223,344				€ 55,836		€ 55,836.00		€ 55,836		€ 37,224.00	€ 223,344	€ 0	€ 223,344.00	external future funding	
	Erasmus Universiteit Rotterdam	postdoc	1.00	1-9-2022	31-8-2025		€	7,360	€ 264,960	€	29,440.00		€ 88,320		€ 88,320.00		€ 58,980			€ 264,960	€ 0	€ 264,960.00	external future funding	
	Erasmus Universiteit Rotterdam	postdoc	1.00	1-9-2023	31-8-2026		€	7,360	€ 264,960				€ 88,320		€ 88,320.00		€ 58,980		€ 58,980.00	€ 264,960	€ 0	€ 264,960.00	external future funding	
	Erasmus Universiteit Rotterdam	postdoc	1.00	1-9-2024	31-8-2027		€	7,360	€ 264,960				€ 29,440		€ 29,440.00		€ 88,320		€ 88,320.00	€ 58,980.00	€ 264,960	€ 0	€ 264,960.00	external future funding
	Erasmus Universiteit Rotterdam	technician	1.00	1-9-2022	31-8-2024		€	6,413	€ 153,912	€	25,652.00		€ 76,956.00		€ 51,304						€ 153,912	€ 0	€ 153,912.00	existing Horizon Europe grant STARGATE
	Totaal							€ 4,099,099	€ 152,263	€ 36,834	€ 444,780	€ 112,069	€ 502,209	€ 142,140	€ 409,782	€ 155,926	€ 283,841	€ 75,756	€ 96,104	€ 50,504	Totaal loonkosten	€ 2,462,208		
2.	Kosten van materialen en hulpmiddelen:						2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner			
	Partner die de kosten maakt	Materiaal / hulpmiddel					Kosten																	
	Onderzoeksorganisaties																							
	TU Delft	Science projects with primary/secondary school students					€	10,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	workshops with industry					€	10,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	workshops for early career researchers					€	10,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	seminars on environment and health					€	50,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	Team building events for Flagship					€	50,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	Educational workshops for students					€	20,000												€ 0	€ 0	€ 0.00	convergence	
	TU Delft	material costs					€	200,000					€ 37,297.00		50000		€ 50,000.00	€ 50,000.00		€ 50,000	€ 137,297	€ 187,297.00	external future funding + 12703 euro convergence	
	Erasmus MC	material costs					€	200,000					€ 37,297.00		50000		€ 50,000.00	€ 50,000.00		€ 50,000	€ 137,297	€ 187,297.00	external future funding + 12703 euro convergence	
	Erasmus Universiteit Rotterdam	material costs					€	100,000									€ 47,296.00	€ 100,000.00	€ 40,000.00	€ 0	€ 87,296	€ 174,296.00	external future funding + 12704 euro convergence	
	Totaal							€ 650,000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 74,594	€ 0	€ 100,000	€ 0	€ 147,296	€ 100,000	€ 40,000	sten materialen en hulpmiddelen	€ 461,890		

3.	Kosten van gebruik van machines en apparatuur															2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	external future funding															
	Partner die de kosten maakt	Machine / apparaatuur	Prijs per gebruikseenheid	Gebruiks-eenheden	Kosten																																									
	Onderzoeksorganisaties																																													
	TU Delft	Handheld Spectrometer	€ 50,000	1	€ 50,000							€ 50,000.00											€ 0	€ 50,000	€ 50,000.00																					
				Totaal	€ 50,000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 50,000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	I kosten machines en apparatuur	€ 50,000																				
4.	Aan derden verschuldigde kosten															2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	convergence convergence															
	Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten																																										
	Onderzoeksorganisaties																																													
	TU Delft	support for writing collaborative grant proposals		€ 10,000																			€ 0	€ 0	€ 0.00																					
	TU Delft	website building		€ 5,000																			€ 0	€ 0	€ 0.00																					
			Totaal	€ 15,000	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	Totaal kosten derden	€ 0																				
5.	Publicatie, reis- en verblijfkosten															2022		2023		2024		2025		2026		2027		Totaal		Totaal per partner	convergence convergence convergence convergence convergence convergence															
	Partner die de kosten maakt	Naam derde, omschrijving kosten		Kosten																																										
	Onderzoeksorganisaties																																													
	TU Delft	Conferences		€ 33,333																			€ 0	€ 0	€ 0.00																					
	Erasmus MC	Conferences		€ 33,333																			€ 0	€ 0	€ 0.00																					
	Erasmus Universiteit Rotterdam	Conferences		€ 33,333																			€ 0	€ 0	€ 0.00																					
	TU Delft	Open access publications		€ 20,000																			€ 0	€ 0	€ 0.00																					
	Erasmus MC	Open access publications		€ 20,000																			€ 0	€ 0	€ 0.00																					
	Erasmus Universiteit Rotterdam	Open access publications		€ 20,000																			€ 0	€ 0	€ 0.00																					
				Totaal	€ 159,999	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	€ 0	Totaal kosten derden	€ 0																			
6.	Totale kosten															€ 4,974,098		€ 182,263		€ 38,834		€ 444,780		€ 112,069		€ 502,209		€ 266,734		€ 409,782		€ 255,926		€ 283,841		€ 223,052		€ 196,104		€ 90,504				€ 2,974,098		
	Total convergence funding															€ 2,000,000																														Total external funding

Return address: P.O. Box 80015, 3508 TA Utrecht, The Netherlands

Dr. A. Los
Institute for Housing and Urban Development Studies (IHS)
Erasmus University Rotterdam
PO Box 1935
3000 BX Rotterdam

Subject

Letter of Support 'Decoding Real-Time, Personalized Health Impact of Climate Change and Pollution'

TNO would like to indicate our support to the Proposal '**Real-Time, Personalized Health Impact of Climate Change and Pollution**' as a Convergence of Health and Technology Flagship program.

The program aims at "tailored, precise interventions for all ages, ethnicities, and nationalities, due to our focus on flexible, personalized designs and cost-effectiveness, enabling "individual-tailored precise medical treatment for all". Exposure to air quality is an important health risk factor that will be taken into account in these interventions. The use of high-quality air quality sensor data and personal monitoring in this program, and finding ways to find complementary information and/or generalisation of the results by air quality modelling is an aspect of the program that is innovative and matches the roadmap of TNO on health and environment. The program offers TNO possibilities for collaborations within the Flagship program and joint projects as spin-off of the Flagship program. This would not only enhance the knowledge base for all parties involved, and their ability to compete to further funding, but would also clearly contribute to TNO's mission to create innovations that boost the well-being of society in a sustainable way.

More detailed quantification of the health risks is becoming within reach by the increased data availability and air quality modelling capacity. Integration of personal monitoring data with other data streams is a field of research that is growing rapidly with many aspects still to be tackled. TNO has been active in the fields of sensor development, air quality monitoring and forecasting of air quality, and data assimilation of ground-based and satellite data for more than 20 years. TNO's air quality model LOTOS-EUROS, which is available as an open source model, would be a valuable tool that can be used for the project as a model that is accepted in the Netherlands and Europe (and beyond). Also of high-resolution modelling and AI techniques are used at TNO for air quality and climate applications. The use of sensor data and personalized exposure is currently one of the focal points of TNO's research programs. Exchange of knowledge and where

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Date

3 March 2022

Our reference

100343912

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The General Terms and Conditions for commissions to TNO, as filed with the Registry of the District Court in the Hague and with the Chamber of Commerce and Industry in The Hague, shall apply to all commissions to TNO.

Our General Terms and Conditions are also available on our website www.tno.nl.

A copy will be sent upon request.

possible integration of knowledge and support in quantifying exposure by modelling and sensor integration would be TNO's contribution to the Flagship program when granted.

The proposed project is aligned with TNO's ambitions and expertise and we expect a positive impact on the following

- ✓ Innovations in generation and use of exposure assessment data, in particular of personalized measurements
- ✓ Better understanding of the environmental health risk factors
- ✓ Providing data that can stimulate healthy behaviour and/or make a solid foundation for the creation of healthier environments

Please accept this letter as formal recognition of the value of this proposal in helping TNO maintain and develop our role in creating a sustainable and healthy society.

Yours faithfully,

 Valid Signed by peter wolfs
on 2022-03-09 21:43:20

Ir. P. (Peter) Wolfs
Director Market Unit Circular Economy & Environment

Date

3 March 2022

Our reference

100343912

Page

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To whom it may concern

Subject: Pollution and Health | Letter of intent

Amsterdam, 7 March 2022

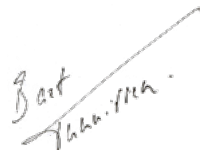
With this letter I declare, on behalf of **Waag Futurelab**, our intention to support the **Convergence Flagship project “Pollution & Health”** that aims at mitigating chronic diseases, including cancer, cardiovascular, neurological, and autoimmune diseases, caused by exposure to environmental degradation, such as climate change and air pollution.

Current research on the links between health and pollution-related environmental stressors are limited to long-term trends and large-scale statistics, restricting the ability of clinicians to provide effective, timely, personalized prevention and treatment strategies, while disproportionally disadvantaging populations that are subjected to racism, poverty, sexism, and heterosexism.

To bridge personal health and environmental stress, especially for the disadvantaged populations, Waag intends to contribute to workshops and personal pollution exposure data products with our knowhow and open data, related to citizen science and citizen initiatives.

Having worked with IHS on related projects before, Waag strongly supports the Convergence Flagship project “Pollution & Health” in this effort as it also effects the work Waag does in this context, to engage European citizens and communities of practice in a social and economic innovative Europe.

Kind regards,



Bart Tunnissen

Director project development and legal affairs

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KvK Amsterdam 41214445

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Address Nieuwmarkt 4

1012CR Amsterdam

The Netherlands

Date March 14, 2022
Our reference TNW 22.128
Contact person Nikky Antonise
Telephone +31 (0)15 27 88087
E-mail d.j.a.antonise@tudelft.nl
Subject Letter of Support



Faculty of Applied Sciences
Chemical Engineering

Visiting address
Lorentzweg 1
2628 CJ Delft
Postal address
P.O. Box 5046
2600 GA Delft
The Netherlands

Dear coordinator of the Convergence Call for the Flagship Programs,

The head of department of the Chemical Engineering department within the faculty of Applied Sciences of Delft University of Technology supports the Flagship Programs application with title "Personalized, Real-Time Health Impact of Climate Change and Pollution".

The main lead of the application is Dr. A. (Alina) Rwei, assistant professor at Chemical Engineering. Dr. Rwei currently has a tenure track position until 1 January 2028.

Yours sincerely,

A handwritten signature in black ink, appearing to be 'B. Dam'.

Prof. dr. B. Dam
Head of Department Chemical Engineering